The boundaries of reliable reasoning

Priority: mathematical feasibility and optimized navigation from a question to a supported answer.

Core ask. Determine the theoretical limits of the proposed knowledge-graph architecture and the best practical design using existing tools. Specify every transition: question → interpretation → evidence retrieval → reasoning → answer or explicit gap. Define what can be guaranteed, under which assumptions and resource budgets.

01 Define the optimization problem

Optimization + decision theory

Risk / open question
What makes a route "best": relevance, evidence coverage, scientific validity, latency or cost? The shortest graph path need not give the best answer.

What we need to resolve
Specify admissible answers, measurable objectives and constraints. Define tradeoffs between quality and resources. Distinguish exact solutions, approximations and heuristics; state where optimality can actually be proved.

02 Preserve meaning in the knowledge layer

Knowledge modeling + scientific NLP

Risk / open question
Can extraction, merging or segmentation discard evidence needed later? A graph does not automatically preserve all information in its source documents.

What we need to resolve
Define claims, identities, conditions and provenance; retain source evidence. Compare section-aware text, graph and hybrid representations. Test whether relevant questions remain answerable after transformation; identify information loss and recovery paths.

03 Bound the search space

Graph algorithms + complexity

Risk / open question
Unrestricted path enumeration can grow exponentially with depth. Disconnected or missing links can hide evidence; aggressive pruning can remove useful routes.

What we need to resolve
Analyze runtime and memory under explicit graph/query assumptions. Compare indexed search, bounded traversal and adaptive expansion. Quantify pruning versus recall; benchmark against exhaustive search where tractable. Specify depth, fan-out and stopping limits.

04 Make the answer path inspectable

Query planning + reasoning systems

Risk / open question
A neural network does not inherently navigate the complete graph or identify every missing premise. An early retrieval error can derail the final answer.

What we need to resolve
Design an explicit planner: resolve intent, decompose the question, retrieve, check evidence coverage, expand if needed, then synthesize. Define contracts for every stage. Trace subquestions to evidence; handle ambiguity, conflicts and failed retrieval.

05 State guarantees and failure boundaries

Formal methods + uncertainty

Risk / open question
Finite evidence cannot answer every question. A missing link does not prove no relationship exists; inventing a bridge risks hallucination.

What we need to resolve
State assumptions for soundness, completeness and termination. Separate observed facts, deductions and hypotheses. Require clarification, partial answers or abstention when support is missing. Distinguish unresolved science from pipeline failures.

06 Select tools through controlled tests

ML evaluation + systems engineering

Risk / open question
Model capability alone cannot repair lost evidence or poor search. More graph nodes, tokens or GPU capacity do not establish better reasoning.

What we need to resolve
Compare model roles and text, vector and graph/hybrid retrieval on held-out expert tasks. Measure evidence coverage, supported answers, abstention, latency and cost. Test missing evidence and growth; choose the simplest design meeting agreed bounds.

Requested expert outcome A mathematical problem specification; complexity and information-loss analysis; an end-to-end retrieval/reasoning design with explicit assumptions, failure handling and resource bounds; and a reproducible benchmark. Prioritize graph algorithms, optimization and retrieval expertise, supported by scientific and ML review.

Scope: expert review of the proposed architecture, not a claim of universal optimality or completeness. Working basis: MeatCODE engineered RAG architecture, curation/scaling proposal and evaluation protocol.